Experiment 5
Archimedes' Principle

SAFETY

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances** attempt to repair any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

WATER AND ELECTRICITY DO NOT MIX

In this experiment you must be aware of the danger of using electrical equipment near water. If any container breaks, please turn off the electrical equipment at the power point and immediately consult your demonstrator.

Ensure that you **use the drip trays provided**, to minimise the possibility of any spillage reaching the electrical apparatus.

Outline of Laboratory Exercise

In **Part A** you will observe the principle of buoyancy (Archimedes' Principle) in action. Using your theoretical knowledge of buoyancy you will predict and test the apparent weight of a fully submerged object.

In **Part B** you will determine the density of an unknown solution by observing the flotation of an object of known density. (Text reference – chapter 14-5)

✓ **Pre-Lab Exercises:** *Read the laboratory exercise and the appropriate section of your textbook (detailing buoyancy and Archimedes' Principle) then complete the online pre-lab questions.*

Before beginning the lab please open up: http://hyperphysics.phy-astr.gsu.edu/hbase/pbuoy.html and read the section on buoyancy.

Learning Goals

- To use a known model to determine a quantity of an unknown material, hence identifying the material.
- To minimise errors through the careful measurement of experimental data.
- To understand the ideas of buoyancy and density.

Introduction

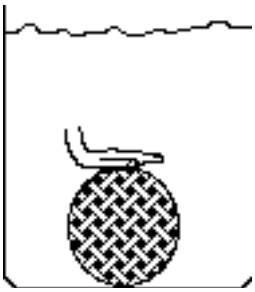
When part (or all) of an object is submerged in a fluid, the fluid exerts an upward force on the object (against the pull of gravity). The discovery of this **buoyancy force** is credited to Archimedes (287–212 BC), from whom we have **Archimedes' Principle**:

> *The buoyant force on an object immersed in a fluid is equal to the weight of the volume of fluid displaced by that object.*

This can be shown by considering the case of an 'object' immersed in the fluid is actually made of the fluid itself (e.g. a particular volume of water immersed within a larger body of water). Obviously, any volume of water must experience a pull of gravity due to its own weight. However, we know that water floats on water – it does not sink through itself! Since our volume of water is not accelerating downwards, a buoyancy force must somehow exist to exactly cancel the gravitational force of its weight. This buoyancy force is due to the pressure of the rest of the water surrounding and below our volume.

It should now be clear that if our volume of water (or any other fluid) is replaced by another object that takes up the same volume, the buoyancy force due to the fluid does not go away. Therefore, any object in a fluid must experience the same buoyancy force that the fluid itself experiences – a force to cancel out the weight of the fluid of that volume.

For example, consider the following situation. A ball is held under water:



A ball of mass $\mathbf{M_{ball}}$ and volume $\mathbf{V_{ball}}$ will experience a gravitational force equal to $\mathbf{M_{ball}\, g}$ (directed downwards). Submerged in water, the ball also experiences a buoyant force equal to the weight of displaced water (directed upwards). The weight of displaced water is the weight of water that would be there if the ball was not – i.e. the volume of the ball multiplied by the density of water.

The object therefore experiences a net force of gravity minus buoyancy:

$$F_{net} = M_{ball}g - \rho_{water}V_{ball}g$$
$$F_{net} = \rho_{ball}V_{ball}g - \rho_{water}V_{ball}g$$
$$F_{net} = \left(\rho_{ball} - \rho_{water}\right)V_{ball}g$$

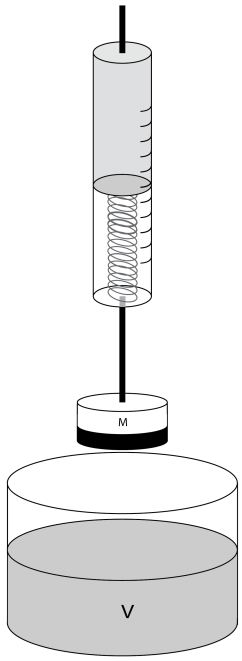
As a result of this, the ball **appears to weigh less** underwater than out of the water – the force it feels is as if the ball is less dense. This is why people feel 'lighter' when they swim. If the density of the ball is less than the density of the water, then the net force is negative and the ball will float upwards, until it is only partially submerged. (As it goes from being completely submerged to only partially submerged, the buoyancy decreases, because the volume of fluid displaced is less – only the parts of the object that are underwater are actually **displacing** fluid.)

According to legend, Archimedes discovered this principle when taking a bath, and was so excited at his discovery that he ran naked through the streets, shouting 'Eureka!' (Greek for 'I have found it!'). Archimedes was considering how to determine if a king's crown was pure gold or not. If the crown is weighed first in air and then underwater, Archimedes' Principle can be used to determine its relative density – i.e. whether or not the crown has the same density as pure gold. And of course, Archimedes' Principle is especially useful for determining whether ships will sink or float.

Part A *Density of an Irregular Object*

In this experiment we will use Archimedes' principle to deduce the density of an irregular object, fully submerged in a fluid of known density. To do so we will use its actual mass *M*, and its apparent mass when fully submerged in the liquid $M_{apparent}$.

Consider the experimental set up below:



Procedure

- Using the scales determine the mass of the given masses. Confirm this with the triple-beam balance in the lab. Why would we need to check these values?
- How many masses you will submerge in order to obtain a linear plot?
- Starting with the lightest mass, submerge the masses while attached to the scales. Record the apparent mass.
- Compare the weight of the masses both out of the water and submerged.
- Write down the formula derived in the pre-lab for ΔM. Within your group, discuss how you will linearise the data and oōtain a linear plot in order to calculate the density of the mass.

Analysis

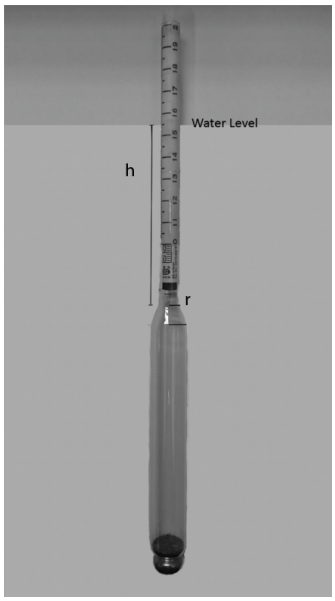
tlot your linearised data, calculate the density of the material submerged. Compare it to the densities listed below. Within error bars is it possible to determine the material? What if we tested Gold?

Material	Aluminium	Brass	Copper	Gold	Iron	Lead	Nickel	Steel
Density (kg/m^3)	2712	8400-8730	8940	19320	7850	11340	8908	7850

Part B *Density of an unknown solution*

In this section you will determine the density of an unknown solution ρ_{sol} (i.e. water with something unknown dissolved in it) by comparing its buoyancy force to the buoyancy of pure water.

Recall that a balanced floating object is at rest: its gravitational weight is exactly balanced by the buoyancy force. Consider a **hydrometer** tube, partially submerged in water as shown:



Some useful data regarding your hydrometers:

$$M_{hyd} = 32.31 \pm 0.1 \text{ g}, \quad r = 4.3 \pm 0.2 \text{ mm.}$$

where ***r*** is the radius of the narrow part of the hydrometer, i.e. the part at water level while floating.

Procedure

- Carefully place the hydrometer in the container of **distilled water**, making sure that it does not sink.
- Let the hydrometer tube come to equilibrium (this should only take about 20 seconds).
- Record the water level (according to the hydrometer's scale on its side).
- Now place the hydrometer in the container of **unknown solution**. Measure the water level of the floating hydrometer as before.

Analysis

- Using your results, calculate the density of the unknown solution. (Assume that the density of pure water is 1.00 g cm^{-3}.) Ask your demonstrator for the actual density of the unknown solution and compare with your measured value.
- How well do the measured and 'actual' values agree?
- If they do not agree, what could be the cause of the discrepancy? Consider and explain.

Conclusion

Summarise your findings and how well they compare with the theory.